The National Institute of Engineering, Mysuru

Autonomous Institute, Affliated to VTU

North Campus No.50 (Part), Koorgalli village, Hootagalli Industrial Area, Mysuru-570018

Karnataka

Department of Computer Science and Engineering

Laboratory Record

For

Full Stack Development – BCSL504



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INDEX

Expt. No	Experiment Title	Page No
1	Design a registration form containing: 1. Username 2. Email 3. Mobile Number 4. Password 5. Confirm Password 6. Branch Selection (dropdown) 7. Upload Profile Picture 8. Register and Reset buttons Implement JavaScript program that validates: i. Username must be at least 6 characters. ii. Email must follow correct format. iii. Mobile number must contain exactly 10 digits. iv. Password must include letters and numbers. v. Confirm Password must match. vi. Error messages should appear next to each field. After successful validation, display the user details in a formatted card using DOM elements.	

Experiment 1

Design a registration form containing:

1. Username
2. Email
3. Mobile Number
4. Password
5. Confirm Password
6. Branch Selection (dropdown)
7. Upload Profile Picture
8. Register and Reset buttons

Implement JavaScript program that validates:

i. Username must be at least 6 characters.
ii. Email must follow correct format.
iii. Mobile number must contain exactly 10 digits.
iv. Password must include letters and numbers.
v. Confirm Password must match.
vi. Error messages should appear next to each field.

After successful validation, display the user details in a formatted card using DOM elements.

Code:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Registration Form</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      background: white;
      display: flex;
      justify-content: center;
      padding: 30px;
    }
    .container {
      width: 420px;
    }
    .form-box, .user-card {
      background: white;
      padding: 25px;
      border-radius: 10px;
      box-shadow: 0 4px 12px rgba(0,0,0,0.15);
    }
    h2 {
      text-align: center;
      color: brown;
    }
    .form-group {
      margin-bottom: 15px;
    }
    label {
      display: block;
      margin-bottom: 5px;
```

```
            font-weight: bold;
        }
        input, select {
            width: 100%;
            padding: 10px;
            box-sizing: border-box;
            border: 1px solid #aaa;
            border-radius: 5px;
        }
        .error {
            color: red;
            font-size: 13px;
            margin-top: 4px;
            display: block;
        }
        .buttons {
            display: flex;
            gap: 10px;
            margin-top: 20px;
        }
        button {
            flex: 1;
            padding: 10px;
            border: none;
            border-radius: 5px;
            cursor: pointer;
            color: white;
            font-size: 15px;
        }
        #registerBtn {
            background: #28a745;
        }
        #resetBtn {
            background: #dc3545;
        }
        .user-card {
            margin-top: 25px;
            display: none;
            text-align: center;
        }
        .user-card img {
            width: 100px;
            height: 100px;
            object-fit: cover;
            border-radius: 50%;
            border: 3px solid #28a745;
            margin-bottom: 15px;
        }
        text-align: left;
            padding: 5px;
            border-bottom: 1px solid #ddd;
        }
    </style>
</head>
<body>
```

```
<div class="container">

  <div class="form-box">
    <h2>Registration Form</h2>

    <form id="registrationForm">

      <div class="form-group">
        <label for="username">Username</label>
        <input type="text" id="username">
        <span class="error" id="usernameError"></span>
      </div>

      <div class="form-group">
        <label for="email">Email</label>
        <input type="text" id="email">
        <span class="error" id="emailError"></span>
      </div>

      <div class="form-group">
        <label for="mobile">Mobile Number</label>
        <input type="text" id="mobile">
        <span class="error" id="mobileError"></span>
      </div>

      <div class="form-group">
        <label for="password">Password</label>
        <input type="password" id="password">
        <span class="error" id="passwordError"></span>
      </div>

      <div class="form-group">
        <label for="confirmPassword">Confirm Password</label>
        <input type="password" id="confirmPassword">
        <span class="error" id="confirmPasswordError"></span>
      </div>

      <div class="form-group">
        <label for="branch">Branch</label>
        <select id="branch">
          <option value="">-- Select Branch --</option>
          <option value="Computer Science">Computer Science</option>
          <option value="Information Science">Information Science</option>
          <option value="Electronics">Electronics</option>
          <option value="Mechanical">Mechanical</option>
          <option value="Civil">Civil</option>
        </select>
        <span class="error" id="branchError"></span>
      </div>

      <div class="form-group">
        <label for="profilePicture">Profile Picture</label>
        <input type="file" id="profilePicture" accept="image/*">
        <span class="error" id="pictureError"></span>
      </div>
```

```
            <div class="buttons">
                <button type="submit" id="registerBtn">Register</button>
                <button type="reset" id="resetBtn">Reset</button>
            </div>

        </form>
    </div>

    <div class="user-card" id="userCard">
        <h2>Registration Successful</h2>

        <img id="cardImage" src="" alt="Profile Picture">

        <p><strong>Username:</strong> <span id="cardUsername"></span></p>
        <p><strong>Email:</strong> <span id="cardEmail"></span></p>
        <p><strong>Mobile:</strong> <span id="cardMobile"></span></p>
        <p><strong>Branch:</strong> <span id="cardBranch"></span></p>
    </div>
</div>
<script>
    const form = document.getElementById("registrationForm");
    form.addEventListener("submit", function(event) {
    event.preventDefault();
        const username = document.getElementById("username").value.trim();
        const email = document.getElementById("email").value.trim();
        const mobile = document.getElementById("mobile").value.trim();
        const password = document.getElementById("password").value;
        const confirmPassword = document.getElementById("confirmPassword").value;
        const branch = document.getElementById("branch").value;
        const picture = document.getElementById("profilePicture").files[0];

        const usernameError = document.getElementById("usernameError");
        const emailError = document.getElementById("emailError");
        const mobileError = document.getElementById("mobileError");
        const passwordError = document.getElementById("passwordError");
        const confirmPasswordError =
            document.getElementById("confirmPasswordError");
        const branchError = document.getElementById("branchError");
        const pictureError = document.getElementById("pictureError");

        usernameError.textContent = "";
        emailError.textContent = "";
        mobileError.textContent = "";
        passwordError.textContent = "";
        confirmPasswordError.textContent = "";
        branchError.textContent = "";
        pictureError.textContent = "";

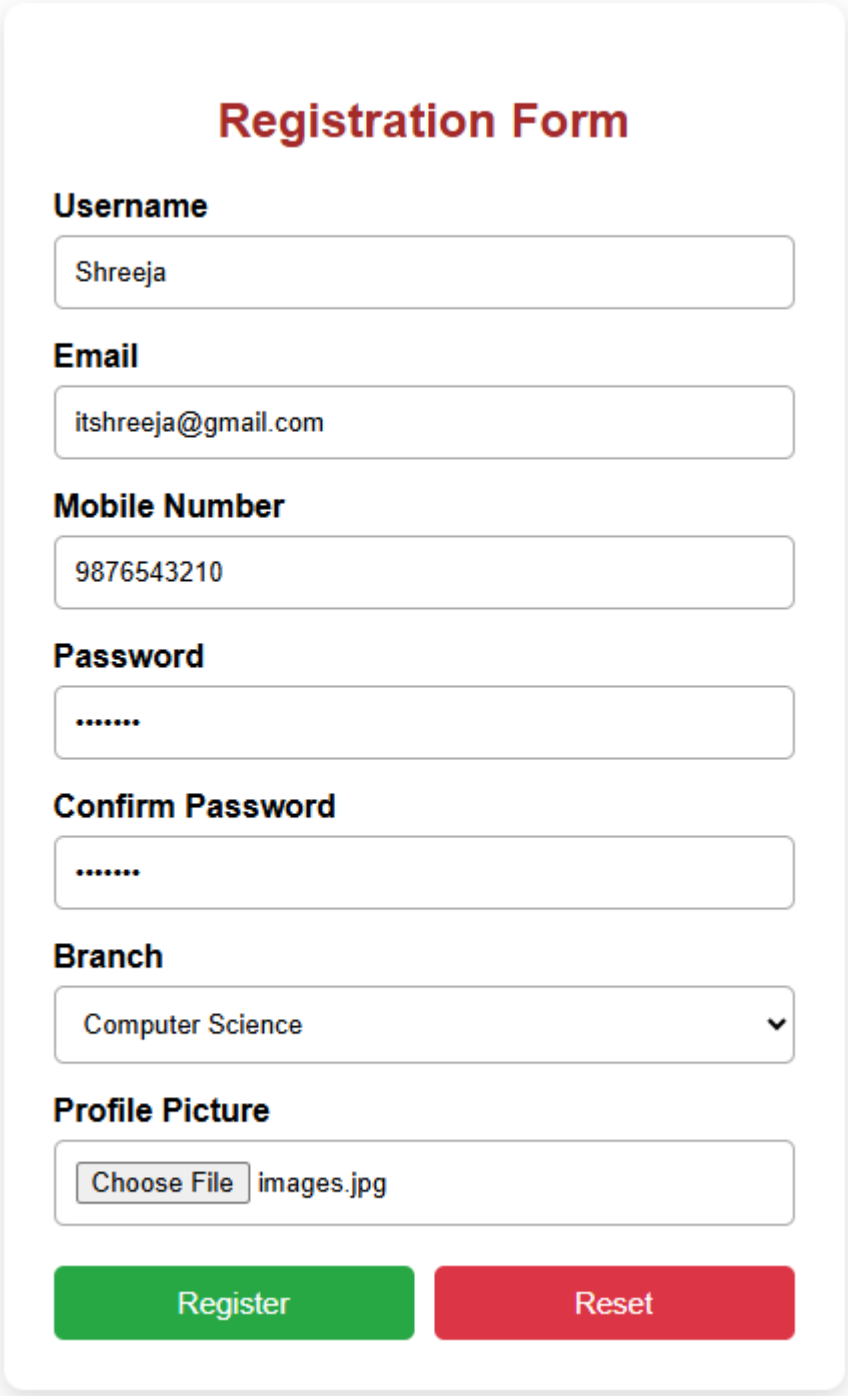
        let valid = true;
        if (username.length < 6) {
            usernameError.textContent =
                "Username must be at least 6 characters.";
            valid = false;
        }
```

```
    const emailPattern =
        /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
    if (!emailPattern.test(email)) {
        emailError.textContent =
            "Please enter a valid email address.";
        valid = false;
    }
    const mobilePattern = /^\d{10}$/;
    if (!mobilePattern.test(mobile)) {
        mobileError.textContent =
            "Mobile number must contain exactly 10 digits.";
        valid = false;
    }
    const passwordPattern =
        /^(?=.*[A-Za-z])(?=.*\d).+$/;

    if (!passwordPattern.test(password)) {
        passwordError.textContent =
            "Password must contain both letters and numbers.";
        valid = false;
    }
    if (password !== confirmPassword) {
        confirmPasswordError.textContent =
            "Passwords do not match.";
        valid = false;
    }
    if (branch === "") {
        branchError.textContent =
            "Please select a branch.";
        valid = false;
    }
    if (!picture) {
        pictureError.textContent =
            "Please upload a profile picture.";
        valid = false;
    }
    if (!valid) {
        return;
    }
    document.getElementById("cardUsername").textContent = username;
    document.getElementById("cardEmail").textContent = email;
    document.getElementById("cardMobile").textContent = mobile;
    document.getElementById("cardBranch").textContent = branch;
    const reader = new FileReader();
    reader.onload = function(e) {
        document.getElementById("cardImage").src = e.target.result;
    };
    reader.readAsDataURL(picture);
    document.getElementById("userCard").style.display = "block";
});
form.addEventListener("reset", function() {
    document.querySelectorAll(".error").forEach(function(error) {
        error.textContent = "";
    });
    document.getElementById("userCard").style.display = "none";
```

```
    });
</script>
</body>
</html>
```

Output:



Registration Successful



Username: Shreeja

Email: itshreeja@gmail.com

Mobile: 9876543210

Branch: Computer Science

Registration Form

Username

Shre

Username must be at least 6 characters.

Email

itshreeja

Please enter a valid email address.

Mobile Number

98

Mobile number must contain exactly 10 digits.

Password

•••••

Password must contain both letters and numbers.

Confirm Password

•••••

Branch

-- Select Branch --

Please select a branch.

Profile Picture

Choose File images.jpg